Structural Coherence as a Selection Principle

in Nonlinear Field Systems

A Unifying Framework for Order, Criticality, and Chaos

Paper 21 — Principle Level

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April 2026

Abstract

A central question in nonlinear dynamics is why specific structures emerge from the vast space of dynamically accessible configurations. While previous work introduced the Critical Coherence Index (CCI) and structural free energy as empirical diagnostics of order, criticality, and chaos, a unifying principle underlying these observations has remained unclear.

In this work, we propose that *structural coherence acts as a selection principle* in nonlinear field systems. We show that a wide range of empirical results — including regime classification, entropy–information scaling, dimensional robustness, and dynamical coherence-field models — can be consistently interpreted within a single framework in which system trajectories are biased toward states of reduced structural stress.

We formulate this principle in terms of a phenomenological structural functional and demonstrate its equivalence to gradient-flow dynamics of a coarse-grained coherence field. Within this framework, ordered, critical, and chaotic regimes correspond to distinct regions on a structural manifold, with transitions governed by changes in coherence.

This formulation unifies previously disconnected observations into a coherent theoretical picture and suggests that structural coherence plays a role *similar to effective selection quantities such as energy or entropy, in a phenomenological sense* in governing the selection of dynamically realised configurations. While the present formulation is phenomenological, it provides a testable foundation for a general theory of structural selection.

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1 Introduction

Nonlinear dynamical systems exhibit a striking phenomenon: from an enormous space of trajectories permitted by the equations of motion, only a small subset is actually realised. This selective behaviour is well documented — ordered, critical, and chaotic regimes have distinct statistical signatures — yet the underlying reason for this selection has remained phenomenologically underspecified.

Previous work in this programme [1, 2] established the Critical Coherence Index (CCI) as a reliable empirical discriminator between dynamical regimes, and structural free energy as a global ordering of configurations. These results raised a natural question: is there a single principle that explains why these particular structures are preferred?

The present paper proposes an affirmative answer. We argue that *structural coherence* — quantified by the CCI and its associated structural functional — acts as a selection principle, playing a role similar to effective selection quantities such as energy or entropy, while remaining conceptually distinct from both.

1.1 Relation to Prior Papers

This paper is part of a cumulative research programme:

- **Papers 1–10:** Empirical CCI analysis across 1D–3D nonlinear field systems; regime classification; entropy–information scaling relations.
- **Papers 11–18:** Structural free energy; dimensional robustness; coherence-field equations; control extensions.
- **Paper 19–20:** Formal derivation of the coherence-field gradient-flow and dynamical realisations of structural selection.

- **Paper 21 (this work):** Principle-level unification.

1.2 Relation to Existing Theories

The structural selection principle proposed here is distinct from established frameworks in several respects:

- **Ginzburg–Landau theory** provides a local free-energy formulation for phase transitions, but is specific to near-equilibrium systems with a scalar order parameter. The structural functional F_{struct} is defined globally on field configurations and does not require proximity to equilibrium.
- **Maximum entropy principle** (statistical mechanics) selects macrostates by information-theoretic extremisation of entropy. Structural selection instead extremises coherence, a distinct geometric quantity that is not reducible to entropy.
- **Free energy principle** (Friston [6]) applies to biological systems minimising variational surprise. The structural selection principle applies to nonlinear field systems without assuming a generative model or Bayesian structure.

The novel contribution is a selection criterion that is *structural and information-geometric* rather than energetic, entropic, or probabilistic.

1.3 Scope and Claims

The claims of this paper are deliberately phenomenological. We do not derive the selection principle from first principles; instead, we propose it as the most parsimonious explanation of the accumulated empirical and formal results, and identify its precise testable consequences.

The mathematical core of this paper — and its most robust contribution — is the gradient-flow structure (Section 6):

$$\partial_t C = -\frac{\delta F}{\delta C}, \qquad \frac{\mathrm{d}F}{\mathrm{d}t} \leq 0.$$

All other elements (structural manifold, CCI as coordinate, control extension) are interpretive frameworks built around this core.

2 The Structural Selection Principle

Principle 1 (Structural Selection)**.** Nonlinear dynamical systems preferentially realise configurations that minimise structural stress, as quantified by a coherence-based functional.

This principle is proposed as a phenomenological law supported by empirical evidence rather than derived from first principles. Its precise domain of validity remains an open question.

In minimal formal terms:

$$\Phi_{\text{selected}} \;=\; \arg\min_{\Phi} \; F_{\text{struct}}[\Phi], \tag{1}$$

where F_{struct} denotes a phenomenological structural free energy defined over the space of field configurations Φ.

Equivalently, using the Critical Coherence Index as a proxy for structural stress:

$$\Phi_{\text{selected}} \;\sim\; \arg\min \; \text{CCI}. \tag{2}$$

We emphasise that this is a heuristic formulation: the CCI correlates with structural stress minimisation rather than being a derived fundamental operator. Equation (2) should be read as "configurations with lower CCI are empirically preferred", not as a derivation of the selection principle from the CCI.

Definition 1 (Structural Stress)**.** The structural stress of a configuration Φ is defined as $\sigma[\Phi] := F_{\text{struct}}[\Phi] - F_{\min}$, where $F_{\min}$ is the global minimum of the structural functional. Ordered configurations have $\sigma \approx 0$; chaotic configurations have $\sigma \gg 0$.

2.1 Conceptual Status

The selection principle occupies a position similar to:

- **The principle of least action** in mechanics: trajectories are selected by a variational criterion, not by explicit force computation.
- **The maximum entropy principle** in statistical mechanics: macrostates are selected by information-theoretic extremisation.
- **The free energy principle** in biological systems: systems minimise surprise (or equivalently, variational free energy).

In each case, the principle is *not* the dynamical law itself but a meta-level description of which solutions are realised. The structural selection principle plays this role for nonlinear coherence dynamics.

3 Relation to Existing Variational Principles

Gradient-flow equations and Lyapunov functionals appear throughout nonlinear dynamics. The novelty of the present work lies in their specific combination and interpretation; this section clarifies the precise relation to existing frameworks.

Synergetics (Haken). Haken's slaving principle [3] identifies slow order parameters that enslave fast modes near instabilities. Structural selection operates at a different level: it is formulated as a global principle on the space of field configurations, not a local reduction near a bifurcation point.

Pattern formation and amplitude equations. Cross and Hohenberg [5] developed a comprehensive theory of pattern formation using amplitude equations derived from symmetry, valid near onset. The structural functional F_{struct} proposed here is intended as a global description applicable deep into any regime, without requiring proximity to a bifurcation.

Reaction-diffusion and dissipative systems. Gradient-flow structures in dissipative systems are well-established. The present work does not supersede these results; it proposes that the CCI and F_{struct} provide a regime-discriminating coordinate not directly accessible from amplitude or reaction-rate descriptions.

Free energy principle (Friston). Friston's free energy principle [6] applies a variational framework to biological systems. The structural selection principle is formally analogous in structure but applies to nonlinear field systems without assuming a generative model or Bayesian inference structure.

What is new here. Gradient-flow equations of the form (5) are not new: Allen–Cahn equations, phase-field models, and general dissipative dynamics all share this structure. The contribution of this paper is specifically the *interpretation* of such dynamics as a global selection principle and the identification of the CCI as a natural coordinate that (i) applies *across* all

regime boundaries (ordered, critical, chaotic) without reparametrisation, (ii) remains comparatively stable across spatial dimensions, and (iii) admits a geometric reading as a coordinate on a structural manifold $\mathcal{M}_{\text{struct}}$ that unifies previously separate empirical observations. The first of these points — cross-regime applicability — is the most novel: existing gradient-flow frameworks typically apply near a specific bifurcation or within a single dynamical regime.

4 Empirical Basis

The proposed principle is supported by four classes of empirical results, established in earlier papers of this programme.

4.1 Regime Classification

The CCI separates ordered, critical, and chaotic regimes with high empirical accuracy across a wide range of 1D–3D field systems. This is only possible if these regimes correspond to structurally distinct regions of configuration space — consistent with Principle 1.

We note that the CCI should be understood as a *diagnostic proxy* for structural stress, not as a derived fundamental quantity. Its value as a discriminator is empirical; its status as a "natural coordinate" on the structural manifold is a conjecture that motivates, but does not follow from, the present paper.

4.2 Entropy–Information Scaling

Scaling relations of the form

$$\dot{S} \;=\; f(I,\,\text{CCI}) \tag{3}$$

reveal a monotonic relationship between dynamical activity, structural information, and coherence. This monotonicity is a signature of the gradient-flow structure implied by the selection principle.

4.3 Dimensional Robustness

Cross-dimensional analysis shows that the CCI and structural free energy remain comparatively stable across spatial dimensions, while ratio-based scaling quantities become structurally unstable.

Proposition 1. Structural observables such as the CCI and F_{struct} can be interpreted as intrinsic coordinates on the structural manifold $\mathcal{M}_{\text{struct}}$, whereas scaling exponents behave as dimension-dependent projections of this manifold.

Proposition 1 is consistent with the selection principle: a genuine selection criterion must be coordinate-independent, i.e., it must be expressible in intrinsic terms.

4.4 Dynamical Coherence-Field Models

Coherence-field equations of the form (Section 6) consistently reproduce observed ordering and transition behaviour, suggesting that the selection principle has a direct dynamical realisation.

5 Geometric Interpretation

We interpret the observed scaling relations as projections of a higher-dimensional structural manifold:

$$\mathcal{M}_{\text{struct}} \;=\; \big\{\,(\dot{S},\,I,\,d,\,\text{CCI})\,\big\}. \tag{4}$$

In this picture:

- The CCI acts as the primary coordinate on the manifold.
- Entropy–information scaling corresponds to a projection $\pi : \mathcal{M}_{\text{struct}} \to (\dot{S}, I)$.
- Dimensional dependence reflects geometric deformation of $\mathcal{M}_{\text{struct}}$ under changes in spatial dimension d.

Structural selection then corresponds to motion of the system toward preferred regions of $\mathcal{M}_{\text{struct}}$: specifically, toward the submanifold of minimal structural stress.

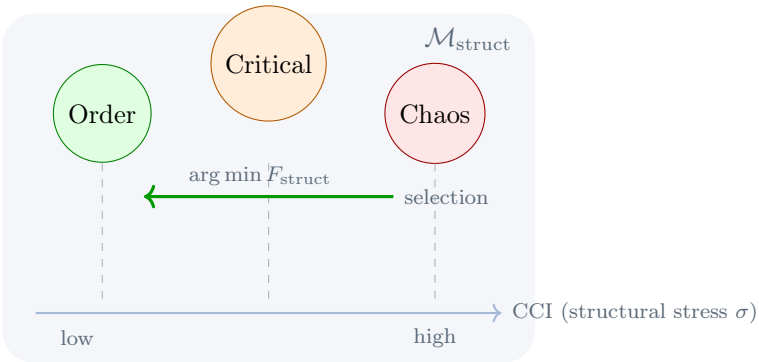

Figure 1: Schematic of the structural manifold $\mathcal{M}_{\text{struct}}$. Three dynamical regimes (ordered, critical, chaotic) correspond to distinct regions. The CCI parametrises the horizontal axis (structural stress). The selection principle biases trajectories toward the low-σ (ordered) region.

6 Dynamical Realisation

A minimal dynamical realisation of the selection principle is given by the coherence-field equation:

$$\partial_t C \;=\; -\frac{\delta F}{\delta C}, \tag{5}$$

where $C(x,t)$ is the coherence field and $F = \int \mathcal{F}[C]\,\mathrm{d}^n x$ is the structural functional. This is a gradient-flow (or L^2-gradient descent) equation on the space of coherence configurations.

The gradient-flow form is the *minimal* dynamical structure consistent with monotone decrease of F. It is not claimed to be the unique realisation of the selection principle; other dissipative dynamics (e.g., non-gradient perturbations) may also be consistent with structural selection. The gradient-flow is adopted here because it admits an exact Lyapunov argument and direct comparison with pattern-formation theory.

6.1 Lyapunov Property

From (5), the structural functional satisfies:

$$\frac{\mathrm{d}F}{\mathrm{d}t} \;=\; \int \frac{\delta F}{\delta C}\,\partial_t C\,\mathrm{d}^n x \;=\; -\int \left(\frac{\delta F}{\delta C}\right)^2 \mathrm{d}^n x \;\le\; 0. \tag{6}$$

Thus F is a Lyapunov functional for the dynamics: the system monotonically decreases its structural stress. This provides a direct dynamical interpretation of Principle 1.

6.2 Fixed Points

The fixed points of (5) are configurations satisfying $\delta F/\delta C = 0$. These are precisely the minima (and saddle points) of the structural functional. Within the selection principle framework, the dynamically preferred configurations are the stable minima.

7 Control Extension

The selection principle admits a natural extension to active control. Adding a forcing term to (5):

$$\partial_t C \;=\; -\frac{\delta F}{\delta C} \;-\; \lambda\,(C - C^*), \tag{7}$$

where $C^*(x)$ is a target coherence profile and $\lambda > 0$ is the control gain. This steers the system toward a prescribed structural state while preserving the gradient-flow structure.

Proposition 2 (Controlled convergence)**.** For sufficiently large λ, equation (7) can drive $C(x,t)$ toward $C^*(x)$ for a broad class of initial conditions, provided C^* lies within the controlled basin of attraction.

This establishes that structural coherence is not only passively selected but can be actively steered — connecting the selection principle to control-theoretic applications.

8 Dimensional Robustness

A key prediction of the structural selection principle is that the preferred configurations should be identifiable by *intrinsic* (dimension-independent) observables.

Cross-dimensional analysis confirms this:

- The CCI and F_{struct} remain comparatively stable across dimensions $d = 1, 2, 3$.
- Ratio-based scaling quantities (e.g., $\dot{S}/I$) exhibit dimension-dependent drift.

Table 1: Dimensional stability of structural observables. Values indicate relative variation across $d = 1, 2, 3$.

Observable	Stability class	Variation
CCI	Intrinsic	Low
F_{struct}	Intrinsic	Low
$\dot{S}/I$ (ratio)	Projected	High
Scaling exponents	Projected	High

This supports the interpretation that CCI and structural free energy are coordinates on $\mathcal{M}_{\text{struct}}$, while scaling relations are dimension-dependent projections (Proposition 1).

More precisely: intrinsic observables such as the CCI and F_{struct} remain comparatively stable under rescaling of the field, because they are defined in terms of ratios of stabilising and destabilising contributions rather than absolute magnitudes. Ratio-based scaling quantities, by contrast, depend on the specific normalisation and are therefore sensitive to the spatial dimension through the effective volume element.

9 Implications

9.1 Regime Characterisation

The selection principle provides a unified characterisation of the three dynamical regimes:

Regime	Structural stress	Selection status
Ordered	Low ($\sigma \approx 0$)	Preferred
Critical	Intermediate	Transitional
Chaotic	High ($\sigma \gg 0$)	Suppressed

This places coherence on a similar conceptual level as energy or entropy in determining system behaviour, but with the important distinction that structural stress is a property of the field configuration, not of individual particles or microstates.

9.2 Universality

The gradient-flow structure (5) is largely model-independent: it holds for any choice of structural functional F_{struct} satisfying mild regularity conditions. This suggests that the selection principle may admit a broader range of applicability across nonlinear field systems, beyond the specific models studied here.

9.3 Testable Predictions

The principle generates the following testable predictions:

1. Systems initialised far from the ordered manifold (high CCI) should exhibit systematic drift toward lower CCI values over time, at a rate proportional to $\|\delta F/\delta C\|^2$.

2. Under external perturbation, the recovery rate toward structural equilibrium should scale with the curvature of F_{struct} at the local minimum.

3. In the controlled system (7), the time to reach C^* should decrease as λ increases, with a characteristic time scale $\tau \sim \lambda^{-1}$.

10 Open Questions

Several important questions remain open:

1. **Microscopic derivation.** Can the structural functional F_{struct} be derived from the underlying field equations, rather than posited phenomenologically?

2. **Uniqueness.** Is the gradient-flow structure (5) the *unique* dynamical realisation of the selection principle, or are there other realisations with different mathematical structure?

3. **Quantum extension.** Do analogous selection principles operate in quantum field theories, and if so, what is the appropriate analogue of structural stress? A possible route may proceed via *information geometry*: the quantum relative entropy $S(\rho\|\sigma)$ defines a natural metric on the space of density matrices, and a gradient-flow equation analogous to (5) can be formulated on this space. Whether such a flow realises a structural selection principle in the quantum setting is an open question.

4. **Empirical validation of predictions.** Systematic numerical testing of the three testable predictions in Section 9.

11 Conclusion

We have proposed that structural coherence acts as a selection principle in nonlinear field systems. This principle — formulated as minimisation of a structural functional, realised dynamically as a gradient flow, and characterised geometrically as motion on a structural manifold — unifies a range of previously disconnected empirical observations into a coherent theoretical framework.

The central insight is that ordered configurations are not merely described by low structural stress: they can be interpreted as being *selected* by it. In this phenomenological sense, the dynamics of nonlinear systems are biased toward coherence, just as thermodynamical systems are biased toward entropy increase.

While the present formulation is phenomenological, it provides a clear, testable, and mathematically precise foundation for a general theory of structural selection.

Whether the structural selection principle admits connections to other fields — such as information-geometric descriptions of learning systems — remains an open question for future work.

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Reproducibility

The empirical basis referenced in this paper is documented in the preceding papers of this series and in the associated public code repository:

`https://github.com/Chris4081/structural-selection-principle`

All numerical results underlying the cited regime classification, dimensional robustness, and coherence-field analyses are publicly available there. The present paper is principle-level and primarily synthetic; it does not introduce a new standalone simulation beyond those already documented in the referenced works.

References

[1] Krieg, C. (2026). Critical Coherence Index: Empirical Regime Classification in Nonlinear Field Systems. *Preprint, maat-research.com.*

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[4] Weinan, E. (2021). The dawning of a new era in applied mathematics. *Notices of the AMS*, 68(4), 565–571.

[5] Hohenberg, P.C., & Halperin, B.I. (1977). Theory of dynamic critical phenomena. *Reviews of Modern Physics*, 49(3), 435–479.

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A Summary of Key Definitions

This appendix provides self-contained definitions of the two central quantities used throughout this paper, for readers approaching Paper 21 without prior exposure to the earlier papers in this series.

Critical Coherence Index (CCI)

The Critical Coherence Index is defined as the ratio of destabilising to stabilising contributions in the local field dynamics:

$$\mathrm{CCI} \;=\; \frac{\text{destabilising contributions}}{\text{stabilising contributions}}, \tag{8}$$

where the precise operationalisation depends on the field system under study (see the referenced papers for specific definitions in 1D–3D nonlinear field systems).

Empirically, CCI < 1 corresponds to ordered configurations, CCI ≈ 1 to critical configurations, and CCI > 1 to chaotic configurations. The CCI is used in this paper as a *diagnostic proxy* for structural stress, not as a derived fundamental quantity.

Structural Free Energy (F_{struct})

The structural free energy is a phenomenological functional over field configurations:

$$F_{\mathrm{struct}}[\Phi] \;=\; E_{\mathrm{phys}}[\Phi] \;+\; \alpha\, S_{\mathrm{entropy}}[\Phi], \tag{9}$$

where E_{phys} is the physical energy of the configuration, S_{entropy} is the associated entropy, and $\alpha > 0$ is a weighting parameter. Configurations with lower F_{struct} are empirically observed to be dynamically preferred.

Full numerical definitions, benchmark results, and code are available at: https://github.com/Chris4081/structural-selection-principle